



3.2 Trigonometry: non-right triangles

Geometry and trigonometry · IB Diploma · Concept guide and original exam-style practice

Understand the concept

For any triangle, the sine rule $a/\sin A = b/\sin B = c/\sin C$ relates sides and opposite angles, and can give two possible solutions (the ambiguous case) when using SSA data. The cosine rule $a^2 = b^2 + c^2 - 2bc \cos A$ finds a missing side (given SAS) or a missing angle (given SSS, rearranged as $\cos A = (b^2 + c^2 - a^2)/(2bc)$). The area of a triangle can be found from two sides and the included angle: $\text{Area} = \frac{1}{2}ab \sin C$.

Worked example: follow the reasoning

Triangle ABC has $a=7$, $b=9$, and angle $C=52^\circ$. Find side c and the area of the triangle, to 2 decimal places.

Solution: $c^2 = 7^2 + 9^2 - 2(7)(9)\cos 52^\circ = 130 - 126(0.6157) = 130 - 77.58 = 52.42$, so $c = \sqrt{52.42} \approx 7.24$.
 $\text{Area} = \frac{1}{2}(7)(9)\sin 52^\circ = 31.5 \times 0.7880 \approx 24.82$.

Quick check · multiple choice

In triangle ABC, $a=10$, $b=14$, angle $A=30^\circ$. Using the sine rule, $\sin B = ?$

- A 0.5
- B 0.7
- C 0.933
- D 1.4

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A surveyor measures a triangular plot PQR: $PQ=120$ m, $PR=95$ m, and angle $QPR=68^\circ$. (a) Find the length QR, to the nearest metre. (b) Find the area of the plot, to the nearest m^2 . (c) Find angle PQR, to the nearest degree. [8]

Final answer: _____



Answers and explanation

Question 1 [8]

(a) $QR^2 = 120^2 + 95^2 - 2(120)(95)\cos 68^\circ = 23425 - 22800(0.3746) \approx 23425 - 8541 = 14884$, so $QR \approx 122$ m. (b) $\text{Area} = \frac{1}{2}(120)(95)\sin 68^\circ = 5700 \times 0.9272 \approx 5285$ m². (c) By the sine rule, $\sin(PQR) = 95 \times \sin 68^\circ / 122.0 = 88.08 / 122.0 \approx 0.7220$, so angle $PQR = \arcsin(0.7220) \approx 46^\circ$.

Quick check answer

B. $\sin B = b \cdot \sin A / a = 14 \times 0.5 / 10 = 0.7$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.